

Integrating Morphology with Gene Expression for Tissue Reassembly

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1 Abstract

Spatial transcriptomics enables high-resolution mapping of gene expression across tissue sections, but current models often overlook the rich morphological context of the tissue. In this project, we extend the LUNA model, a state-of-the-art diffusion-based framework for spatial reconstruction, to incorporate morphology through both cell-level and tissue-level representations. We first embed individual cell images using various image encoders and fuse these embeddings with gene expression data to create multi-modal representations. Additionally, we propose a shape-guided post-processing pipeline that uses tissue morphology to refine spatial predictions. While the integration of cell morphology yielded modest performance gains on our dataset, the shape-guided pipeline proved effective in improving spatial accuracy, particularly in cases where the model struggled with unseen tissue shapes. These results highlight the potential of morphology-aware modeling to enhance spatial transcriptomics analysis, especially in low-data or structurally complex scenarios.

2 Introduction

The human body is an extraordinarily complex, dynamic, and heterogeneous multicellular system - one that no model organism can fully replicate. Understanding its molecular biology has long been a central challenge in biomedical research. This challenge is now being met with the advent of single-cell transcriptomics (ST), which has revolutionized the field by enabling transcriptome analysis within the native spatial context of individual cells [7]. These techniques are crucial for exploring complex gene expressions and their interactions with the cellular microenvironment, offering deep insights into biological mechanisms and disease pathology.

Over the past decade, spatial transcriptomics, powered by both imaging and sequencing-based technologies, has emerged as a transformative tool for mapping genome-wide gene expression and cellular organization within intact tissues. In parallel, single-cell transcriptomics has enabled high-resolution profiling of gene expression by quantifying messenger RNA (mRNA) levels across thousands of genes in individual cells. Together, these advances have made it possible to deconvolute heterogeneous cell populations, reconstruct developmental trajectories, and model transcriptional dynamics - capabilities that were previously obscured in bulk RNA sequencing approaches.

ST methods can be broadly classified into sequencing-based and imaging-based approaches. Sequencing-based technologies such as 10X Visium [14], Slide-seq [10], and Stereo-seq [2] are proficient in capturing whole transcriptomes but offer only spatially aggregated data from multiple cells. Conversely, imaging-based techniques, including Xenium [11], SeqFISH [12], STARmap [16], and MERFISH [15], provide high-resolution data at the single-cell level, but only capture a fragment or a limited set of the transcriptome. This delineation underscores a significant limitation in current ST methods: the lack of capability to collect complete transcriptomes with precise single-cell spatial resolution. It is essential as this would significantly enhance our understanding of cell functions and interactions within varied tissue types.

The earlier work by the MLBio lab introduced LUNA [17], a novel attention-based diffusion model designed to reconstruct spatial cell coordinates from gene expression profiles. By addressing key limitations

of prior methods, LUNA offers a flexible, scalable, and more expressive solution for spatial transcriptomics.

In this work, we extend LUNA into a multi-modal model by integrating tissue morphology as an additional image modality. Cell morphology provides crucial context for interpreting transcriptomic data, as cell shape, size, and structure can significantly influence gene expression patterns. By integrating morphological information with transcriptomic data, we can gain a more comprehensive understanding of cellular behavior and function, including how cells respond to stimuli and the underlying molecular mechanisms driving those responses. We also examine a few limitations of the original LUNA framework and explore how incorporating morphological information can help overcome these challenges and enhance the model’s predictive capabilities.

3 Dataset

We chose a new “Alzheimer’s Disease Mouse Model Brain Coronal Sections” Xenium dataset [11] which includes FFPE brain coronal sections from six mice - three wild-type (normal) and three TgCRND8 (transgenic) mice, that carry a mutant human APP gene to model Alzheimer’s disease. The dataset contains approximately 360,000 cells, with an average of 60,000 cells per donor. Gene expression was profiled using a 347-gene panel, comprising the standard 248-gene Mouse Brain Panel along with a 99-gene custom panel focused on activated microglia, astrocytes, and genes involved in plaque response - key components of Alzheimer’s pathology. The main reason for selecting this dataset was that it combines spatial gene expression data with high-resolution immunofluorescence images, allowing us to extract and analyze individual cell morphologies. Generated using the Xenium platform, this dataset offers a unique opportunity to study Alzheimer’s-related changes at single-cell and spatial resolution.

We split the dataset such that all slices from the *TgCRND8.5-7*(Diseased) donor were used as the test set while the remaining (Normal and Diseased) were used for training.

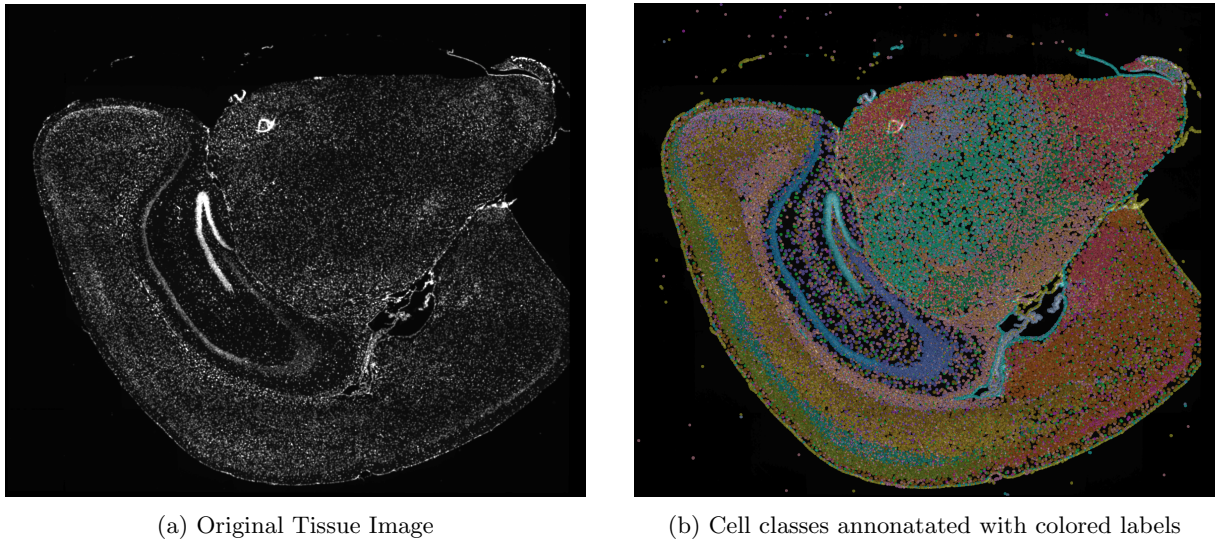


Figure 1: Example of a single tissue slice from a WildType Mouse Brain, containing 58,684 cells

3.1 Data Pre-processing

The data pre-processing pipeline involved two key steps:

1. Merging the 347-gene expression data with the spatial coordinates of each cell
2. Extracting individual cell images from the high resolution tissue image using the Xenium-provided bounding polygons.

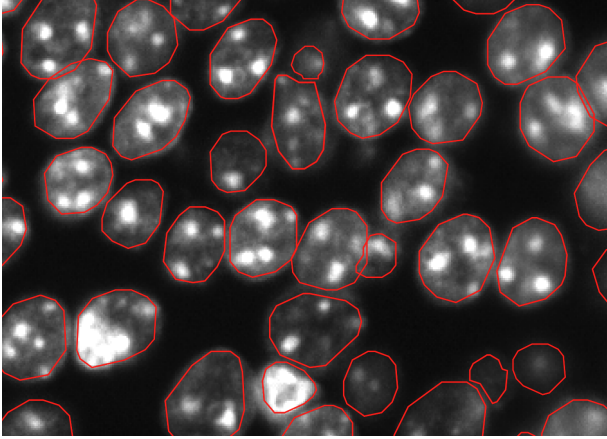


Figure 2: Cell Segmentation based on Bounding boxes provided by Xenium



Figure 3: Example cell Morphology (Wild-Type Samples)

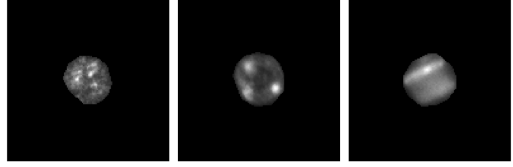


Figure 4: Example cell Morphology (Transgenic Samples)

To prepare the data for analysis, we first addressed the challenge of segmenting individual cells from dense and noisy brain tissue images. The high cellular density led to overlapping or touching nuclei, and low contrast along with uneven illumination made it difficult to accurately detect cell boundaries. We initially explored segmentation techniques such as the Watershed algorithm and Cellpose, both of which performed well generally but struggled with tightly packed nuclei and resulted in under segmentation in dense regions. Ultimately, we used the cell nucleus bounding polygons provided by the Xenium platform [4], which offered reliable and consistent segmentation. These polygons, defined as vertices' coordinates for each nucleus, enabled us to efficiently extract individual high-resolution cell images from the tissue image.

4 Overview: LUNA Model

LUNA is a diffusion-based generative model designed to generate spatial locations of cells from their gene expressions. Using attention-based mechanism, LUNA learns how each individual cell should attend to other cells based on their relevance to a cell's own location and molecular features. During training, LUNA leverages spatial transcriptomics data with ground truth cell locations to learn to generate cell locations conditioned on the gene expressions. During inference, the input to LUNA are the gene expressions of dissociated cells and the model then generates cell locations from pure noise conditioned on the gene expressions, enabling de novo inference of cell locations.

In the training phase, the input to the LUNA is spatial transcriptomics data from single or multiple slices. LUNA learns cell embeddings that capture both the local and global tissue structure using a multi-head self-attention mechanism [19, 20] that enables cells to learn to attend to other cells. The loss in LUNA is defined as an SE(2)-invariant function [35–38], i.e., it is robust to arbitrary rotations, translations and reflections of the predicted locations, only enforcing the preservation of relative distances between cells within each slice.

Training LUNA involves two main steps: (i) a corruption process that iteratively adds noise to the ground truth cell locations, and (ii) a denoising process that generates the ground truth locations from the noise-corrupted locations. During the denoising process, LUNA learns to reverse the corruption process, gradually transitioning from noisier to less noisy cell locations, ultimately reconstructing true cell locations from pure noise. Figure 5 demonstrates the general pipeline of the original LUNA model.

5 Baseline

We set up and evaluated the original LUNA model on our dataset to establish a baseline. This resulted in an average Spearman's rank correlation of 58.9%, surpassing the previously reported best performance of LUNA on the MERFISH-cortex dataset with an average correlation coefficient of 44.8%. These results demonstrate the model's ability to generalize effectively across different datasets. Notably, the predicted cell locations exhibit slight rotational offsets, which is expected due to the SE(2)-invariant [13] (i.e., it is robust to arbitrary rotations,

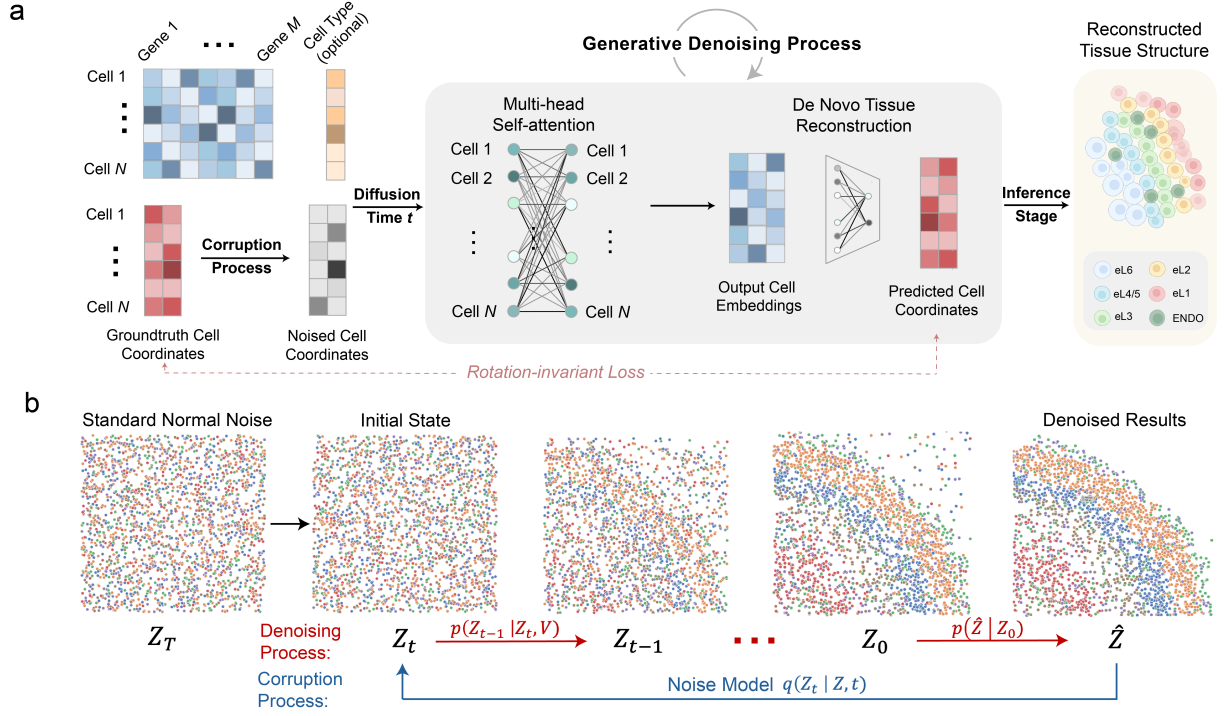


Figure 5: **a** LUNA Framework. **b** The general pipeline of LUNA with diffusion models on spatial reconstruction problem. Here, each scatter represents a cell and they are color-coded by cell class.

translations and reflections of the predicted locations, only enforcing the preservation of relative distances between cells within each slice.) nature of the LUNA model’s architecture.

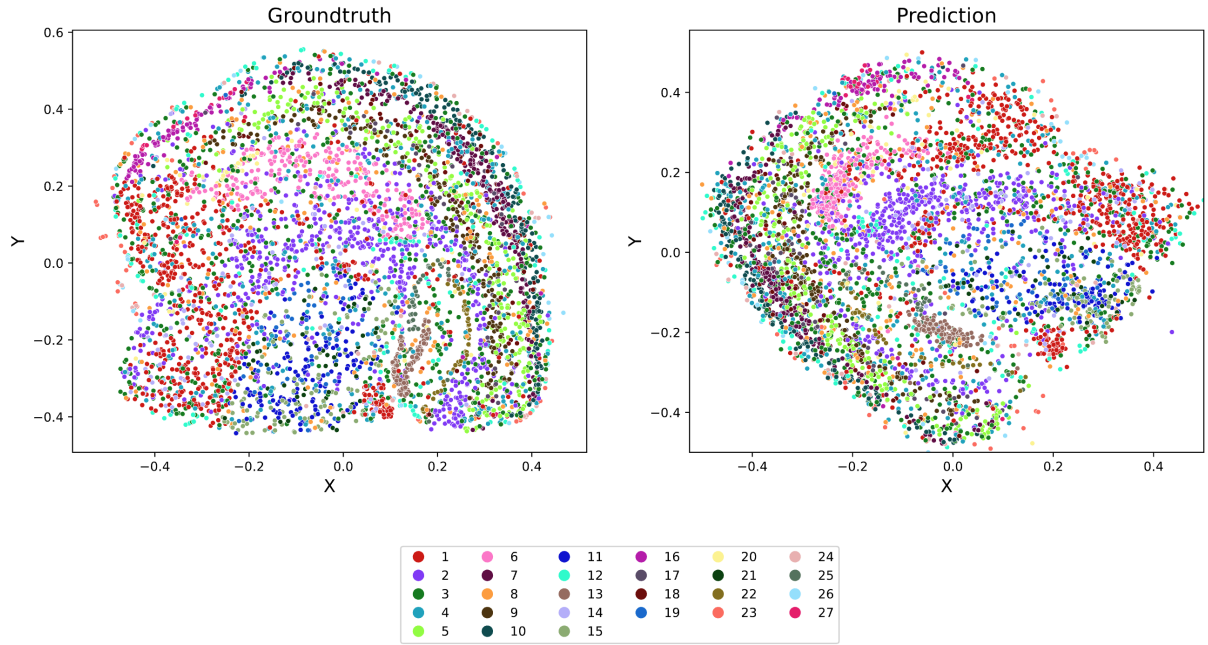


Figure 6: Baseline Result (Original LUNA trained with Xenium data)

6 Motivation: Why add Image Modality?

Combining advanced imaging techniques with spatial transcriptomics offers the advantage of leveraging complementary modalities to gain deeper insights into tissue organization and function. Tissue morphology, especially through high-resolution stained images, can enhance gene expression-based analysis by providing additional context derived from morphology. Additionally, in cases where gene expression data is sparse or absent, morphological features alone can offer valuable insights for downstream tasks.

As outlined in the review by [1], methods that combine spatial transcriptomics with morphological data are broadly categorized into two approaches: translation and integration. In the translation approach, morphological features are transformed into representations that closely correlate with gene expression profiles. This leverages the accessibility and cost-effectiveness of morphological imaging to approximate transcriptomic information, making it particularly valuable in scenarios where gene expression data is limited or unavailable. In contrast, the integration approach combines morphological and transcriptomic data to provide complementary insights. Rather than predicting gene expression, integration captures structural and contextual features that enrich the biological interpretation of spatial data. Integration thus focuses on orthogonal, yet biologically meaningful, information offering a more holistic view of tissue architecture and function.

In our work, we adopted the integration approach, aiming to determine whether morphological features provide additional, non-redundant information beyond gene expression and to improve the capabilities of the original LUNA model. To achieve this, we extracted individual cell images, encoded them using a image encoder to obtain morphology embeddings, and concatenated these with gene expression embeddings prior to input into our transformer-based model.

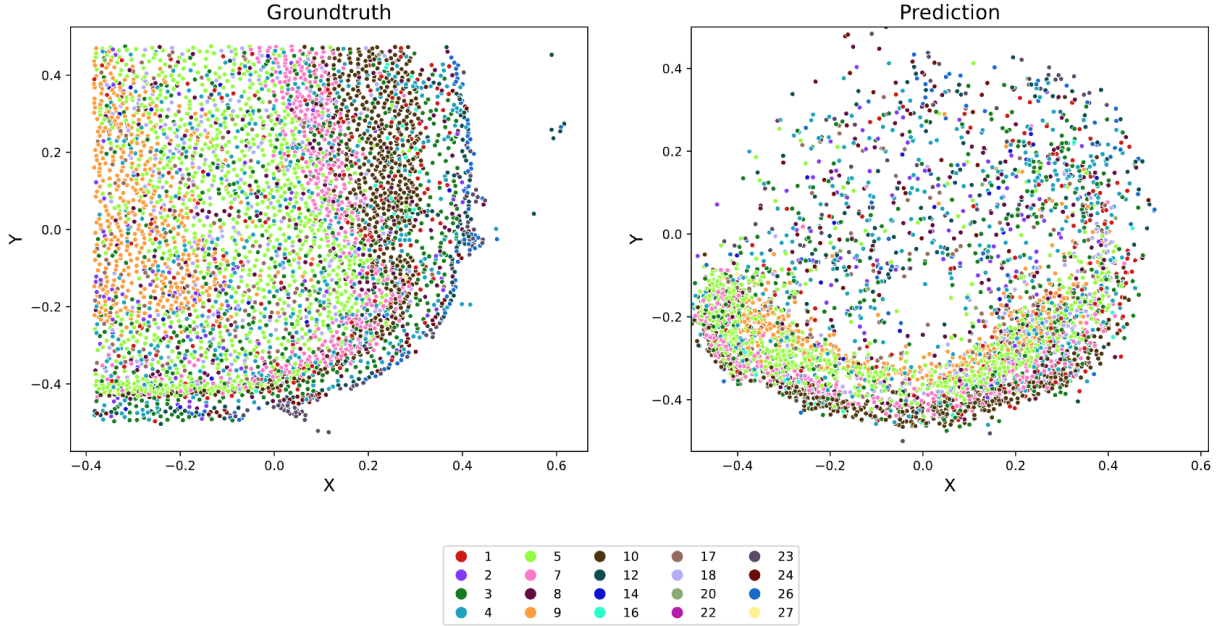


Figure 7: Baseline Result on test slice of unseen shape (Original LUNA model trained with Xenium data)

LUNA employs a mean squared error (MSE) loss on the pairwise distances between the predicted and actual cell locations. This formulation captures the relative spatial arrangement of cells, making the model inherently invariant to global transformations such as rotation and translation. While this loss encourages learning of the overall tissue structure, it does not strongly constrain the global shape of the predicted cell distribution. As a result, the predicted point clouds may still exhibit deviations in tissue morphology. These discrepancies become more pronounced when the model is not exposed to a sufficiently diverse set of tissue shapes during training. For instance, when we evaluated the baseline model on a synthetically generated slice-created by partitioning a single tissue image, it showed diminished performance, with an average Spearman’s rank correlation of 24.7%, as illustrated in Figure 7. To address this limitation, we propose integrating global tissue morphology as an additional guiding signal to better constrain and shape the predicted cell locations.

In summary, we introduce morphological information into LUNA through two key strategies:

1. Integration of morphology features at the individual cell level, by embedding cell images and combining

them with gene expression features.

2. Integration of global tissue morphology to guide the shape of the predicted spatial coordinates.

The following sections provide a detailed explanation of each approach.

7 Part 1: Integrating Cell Morphology with LUNA

We propose an image encoder to extract morphology features from the extracted 128×128 cell images and combine them with gene expression data to obtain joint cell embeddings. These multimodal embeddings aim to capture both transcriptional and morphological information, enabling more informative representations. Figure 8 shows the integration of morphology via cell images into the LUNA model.

We experiment with different image encoders to extract morphology features as shown in figure 9. A simple CNN captures basic texture and shape patterns from grayscale images. The pretrained DINOv2 encoder [8], a vision transformer trained in a self-supervised fashion, captures rich, high-level visual features. The Masked Autoencoder (MAE) [6] approach focuses on reconstructing missing parts of the image, encouraging the model to understand underlying structure and patterns, potentially aligning well with biological priors.

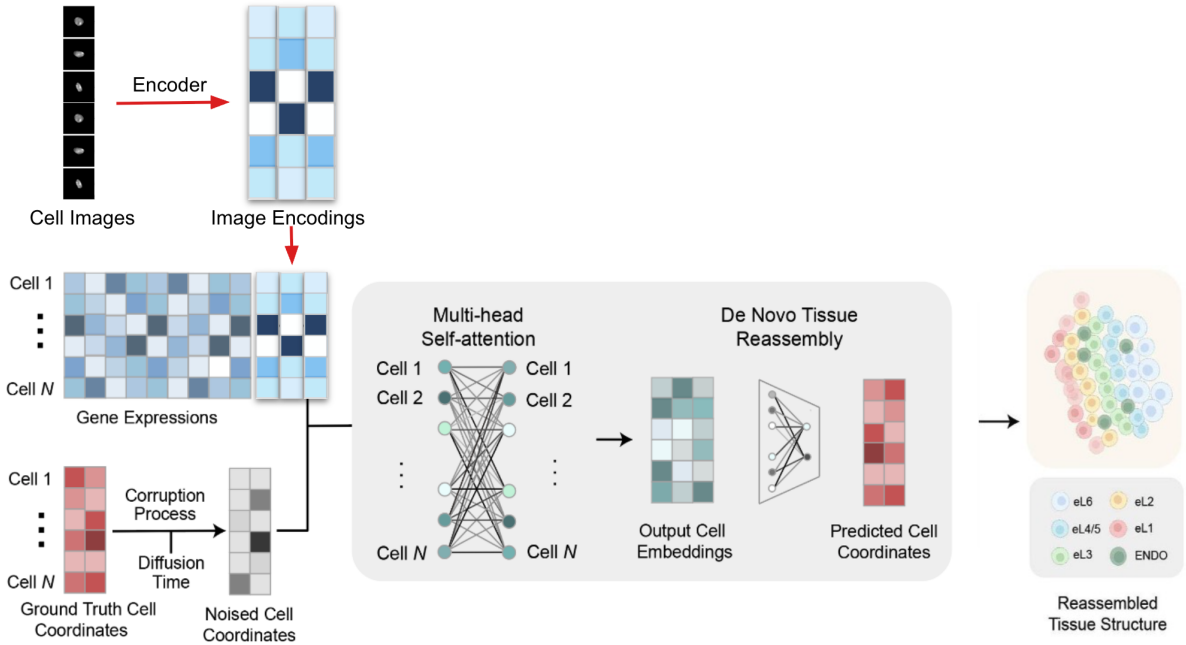


Figure 8: Integration of morphology into the LUNA Model

To obtain a combined representation of each cell, we concatenate the morphology embeddings(extracted from cell images via encoders) with the corresponding gene expression embeddings. This multi-modal representation is subsequently passed through our transformer-based fusion layer that projects the concatenated input into a shared latent space, effectively capturing both morphological context and molecular identity. The fused embeddings are further processed through multiple transformer layers enabling contextual interactions between cells. This allows the model to reason jointly over gene expression and morphology to infer biologically plausible spatial organization.

7.1 Experiments

We used the preprocessed dataset comprising spatial transcriptomics data with known cell coordinates and paired cell morphology images. Each cell is characterized by a gene expression vector and a corresponding 128×128 grayscale image extracted from high-resolution tissue scans. The known spatial coordinates serve as the ground truth for model prediction.

To extract morphology features from 128×128 grayscale cell images, we evaluated three different image encoders:

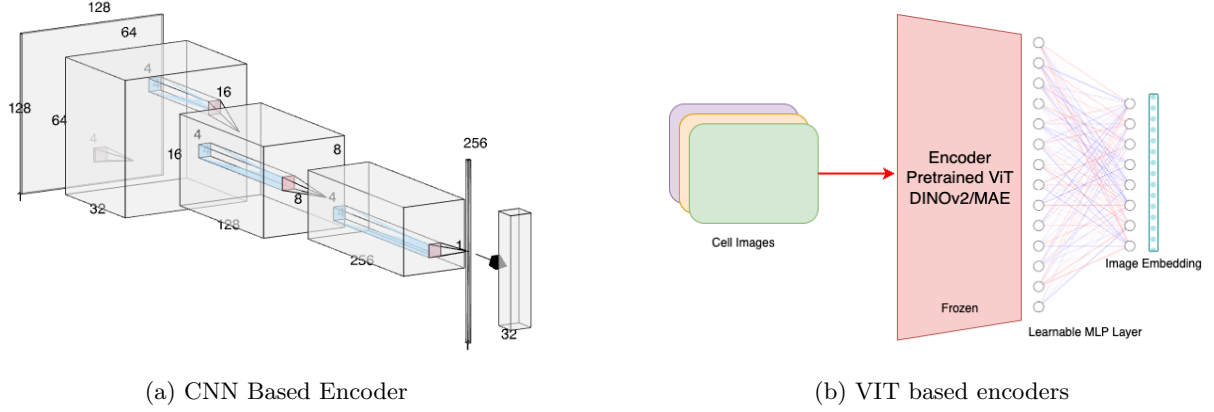


Figure 9: Cell Image Encoders to extract Morphological features

1. **CNN-based Encoder:** A lightweight 4-layer convolutional neural network with GroupNorm and ReLU activations, followed by adaptive average pooling and a linear projection to a fixed-size embedding.
2. **Pretrained DINOv2 Encoder:** We use the DINOv2 ViT-S/14 backbone (384-D output). Grayscale images are first converted to RGB and resized to 224×224 . The model weights are kept frozen, and a trainable MLP projects the output to the target embedding size.
3. **MAE Embeddings:** We use 768-D features extracted from a pretrained Masked Autoencoder (MAE) and pass them through a 2 layer MLP with ReLU activations to obtain the final embedding.

Each of these encoders was integrated into the LUNA model pipeline to assess their impact on spatial reconstruction performance.

We used the same train-test split with all slices from the *TgCRND8.5-7*(Diseased) donor in the test set and the remaining (Normal and Diseased) in the training set. All models were trained using the same optimization settings and transformer backbone for fair comparison. The training was conducted for 500 epochs using the Adam optimizer, a learning rate of 0.005, and a batch size of 4. We used a fixed embedding size of 64 for morphology, after experimenting with other sizes such as 32 and 512. We found that 64 offered a good balance between performance and model stability without significantly increasing computational cost. Morphology encoders were kept frozen during training, except for the CNN variant, which was trained end-to-end. The gene expression features and morphology embeddings were concatenated to form joint multi-modal embeddings, which were fed into the transformer model to predict cell spatial coordinates.

7.2 Results and Discussion

We evaluated all three variants of the model using different image encoders. Table 1 presents a comparison across three key evaluation metrics: Average and Median Spearman’s Rank Correlation (SRC), and Precision. Among the image encoder variants, the vision transformer-based encoders—MAE and DINOv2—demonstrated decent improvements in performance over the baseline. In contrast, the CNN-based encoder yielded results comparable to the baseline, indicating limited benefit from using a lightweight convolutional architecture in this setting.

The superior performance of DINOv2 could be attributed to its self-supervised pretraining on large and diverse natural image datasets, which helps it learn rich and transferable visual features. Despite being trained on non-biological data, its Vision Transformer backbone captures complex textures and spatial patterns in grayscale cell morphology images more effectively. The high-capacity frozen backbone, combined with a learnable MLP head, likely enables task-specific adaptation without overfitting, resulting in more informative multimodal embeddings and better downstream performance.

Overall, the modest improvements observed upon integrating morphology suggests several key insights. First, in this particular setting, morphological features may not provide substantial additional information beyond what is already captured by gene expression profiles. While this may indicate redundancy in the two modalities, it does not rule out the utility of morphology in other contexts, particularly when gene expression is sparse or when high-quality, richly stained tissue images (e.g., DAPI) are available. However, These scenarios need further exploration.

Second, a more detailed investigation is needed to understand the structure and utility of the learned morphology embeddings. Specifically, analyzing the embeddings in the latent space could reveal whether they exhibit biologically meaningful clustering or relationships. If such patterns are absent, it may indicate that morphology does not contribute significantly to spatial inference in the current setup, highlighting the need for either better morphological encoders or different fusion strategies.

Method	SRC(Average)	SRC(Median)	Precision
Baseline	0.589	0.677	0.372
CNN-based Encoder	0.570	0.676	0.361
DINOv2 Encoder	0.624	0.698	0.399
MAE(Masked Auto Encoder)	0.603	0.686	0.390

Table 1: Performance comparison of LUNA morphology-integrated models using different image encoders. SRC stands for Spearman’s Rank Correlation

8 Part 2: Shape-Guidance using global tissue Morphology

One of the key limitations of the original LUNA model is its limited generalizability across tissues with diverse shapes. We hypothesized that this stems from the model being trained on a relatively small dataset, where tissue geometries are often similar. As a result, the model tends to memorize structural patterns seen during training, rather than learning more generalizable spatial principles. This leads to poor performance when predicting cell coordinates for tissue sections with previously unseen or irregular shapes.

Motivated by this limitation, we aimed to improve the model’s ability to generalize by explicitly guiding the shape of the predicted point cloud. Our objective was to ensure that while the model accurately places individual cells, it also preserves the overall tissue geometry in its spatial predictions. We explored methods to incorporate global tissue morphology as a high-level structural cue to address this challenge.

8.1 Method

8.1.1 Synthetic Dataset

To address the limitations posed by a small and shape-biased dataset, we constructed a synthetic dataset derived from the original tissue samples. Specifically, we divided each full tissue slice into 16 slightly-overlapping sub-regions using a 4×4 grid, resulting in approximately 84 individual slices with diverse and irregular shapes as shown in figure 10

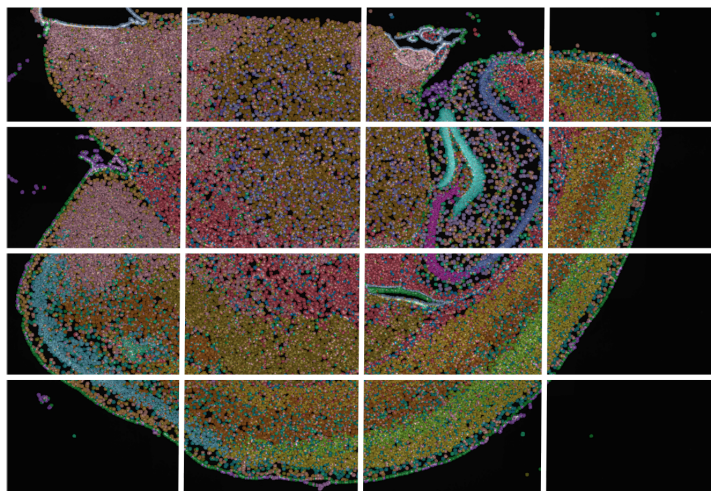


Figure 10: Examples slices from Synthetic Dataset, obtained by dividing tissue into multiple slices

This slicing strategy was designed to introduce greater variability in tissue geometry, enabling the model to encounter a broader range of spatial patterns during training. Correspondingly, we updated the associated gene expression matrices and cell coordinate annotations to reflect their positions within the new cropped regions, ensuring consistency between spatial and molecular data.

Additionally, we designed two distinct train-test splits for the synthetic dataset to assess generalization. In the first setup, all slices from the *TgCRND8-5-7* donor were used exclusively as the test set, with the remaining slices serving as training data. In the second setup, we selected the rightmost slices from each donor as test data, using the remaining slices for training. This ensured that the model encountered unseen tissue shapes during evaluation. By augmenting the dataset in this way, we aimed to encourage better generalization and reduce the model’s reliance on memorized global tissue structures.

8.1.2 Shape-Guidance

To guide the shape of the the predicated cell locations, we explored both in-training and post training methods. In-training approach involved designing a suitable loss function that can capture the shape discrepancy between the tissue image and shape of the predicted cell locations. This shape guided loss is then jointly optimized with the original loss which is the mean squared error (MSE) between the pairwise distance of the actual and predicted cell locations. However, designing such a loss is inherently challenging due to the modality difference, the tissue image is a grayscale image, whereas the predictions form a 2D point cloud. This makes it difficult for the model to learn meaningful patterns or relationships across these modalities. As a result, the model may resort to memorizing tissue shapes seen during training, thereby compromising its generalizability.

The post-training approach seemed more promising. After obtaining the cell location predictions from the model, we applied a transport-based algorithm to deterministically project the cells to the target spatial distribution derived from the tissue image using image processing techniques. This project step aims to align the predicted cell locations with the underlying tissue structure while still preserving the optimization objectives, such as pairwise distance MSE and other evaluation metrics.

- **Spatial distribution from Tissue Image:**

To simulate realistic spatial distributions of cells from a tissue section, we converted a grayscale tissue image into a probabilistic map that guides sampling of cell locations.

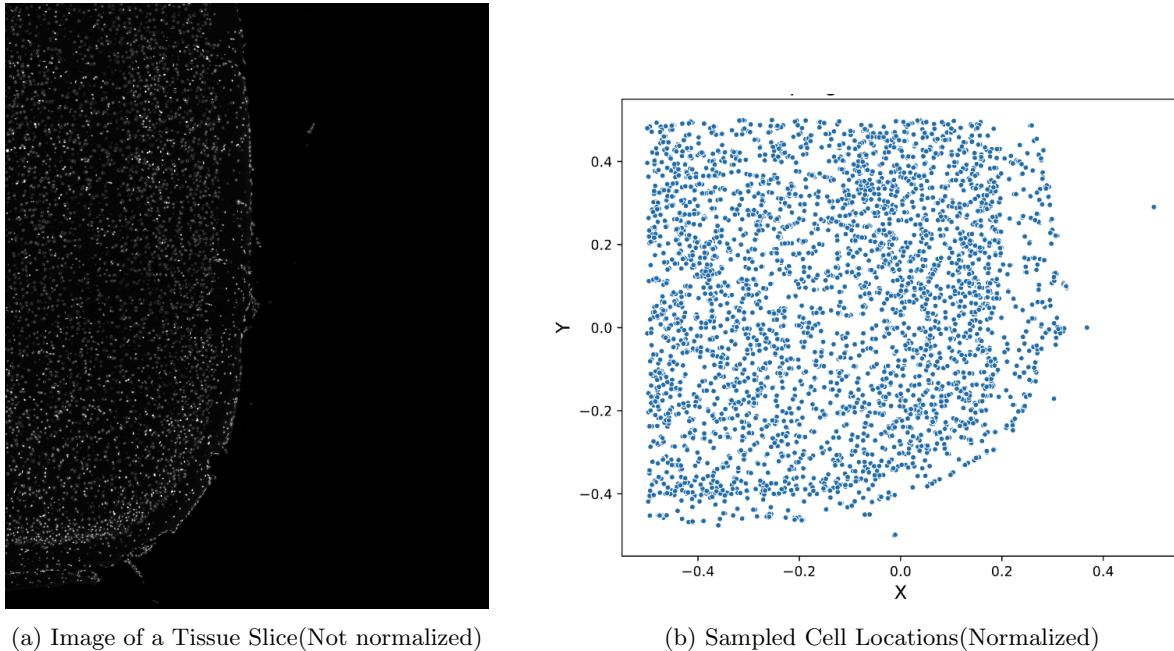


Figure 11: Sampling Cell Locations from the Tissue slice image

We first detect bright peaks in the tissue image, corresponding to likely nuclei, by applying intensity based detection which identifies likely candidate cell locations by finding bright, well-separated regions in the image while also avoiding noise and overlap. These points are used to construct a sparse density map,

which is then smoothed using a Gaussian kernel to spread their influence and produce a continuous spatial density map. The resulting map is then normalized into a valid probability distribution, from which spatial coordinates are sampled. This process yields a synthetic point cloud that mirrors biologically realistic cell distribution patterns in the tissue.

- **Transport-based Algorithm:**

We use a transport-based algorithm to map the predicted cell locations from the model onto a sampled point cloud derived from the tissue image. Since the model’s predictions are invariant to rigid transformations such as rotation, translation, and reflection, a direct correspondence between the two point clouds is not meaningful. To address this, we adopt a two-stage alignment process: first, we apply a rigid transformation (e.g., Procrustes analysis) to coarsely align the overall structure of the predicted and image-derived point clouds. Then, we perform optimal transport to establish a fine-grained, one-to-one correspondence, enabling shape-aware alignment without distorting the local tissue geometry.

1. **Rigid Transformation:** To align the predicted cell coordinates with the ground truth, we perform Procrustes transformation [5], a symmetric rigid transformation, comprising rotation and translation, without scaling or deformation. Both point clouds are first centered by subtracting their respective centroids, isolating their shape. The optimal rotation matrix is then computed using Singular Value Decomposition (SVD) of the cross-covariance matrix between the two centered point sets. This decomposition provides orthogonal matrices that, when combined, yield the rotation minimizing the mean squared error between the aligned and reference points. Finally, the rotated point cloud is translated to match the original centroid of the reference, completing the rigid alignment.
2. **Non-Rigid Transformation(Optimal Transport):** Post rigid alignment, we apply optimal transport [9] to guide the shape using Python’s POT [3] library. This method computes a globally optimal mapping between predicted and reference coordinates by minimizing the total transport cost. Specifically, we use the Hungarian algorithm to achieve a one-to-one correspondence between points based on spatial proximity. Unlike rigid transformations, this approach allows flexible alignment while preserving local structures, making it well-suited for complex tissue geometries. Figure 12 demonstrates Optimal Transport using a toy example.

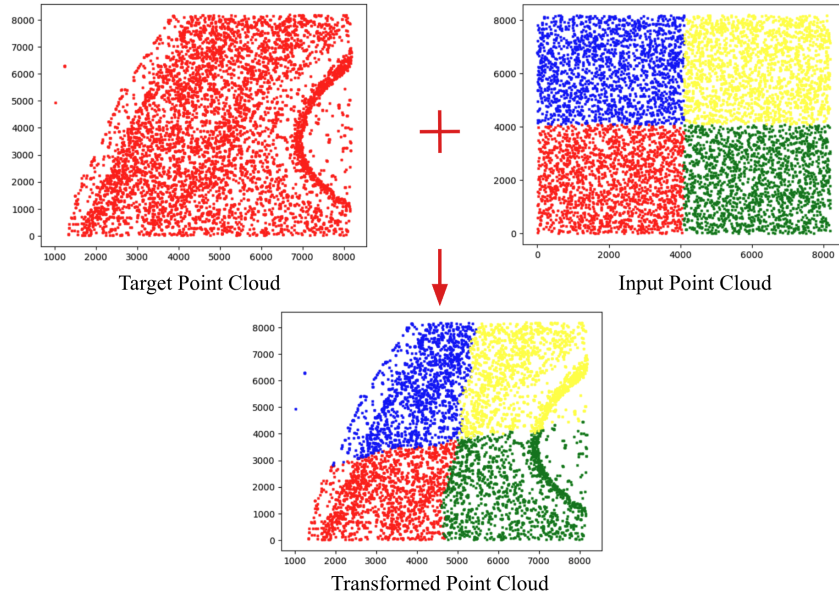


Figure 12: Optimal Transport (Toy Example)

8.2 Experiments

We began our investigation by training the LUNA model on our synthetic, shape-diverse dataset using two carefully designed train-test splits using the same training setup and parameters. Subsequently, we applied our shape-guidance pipeline as a post-processing step to the model’s predictions on the synthetic test set. This allowed us to evaluate how well the pipeline could refine and align the predicted cell distributions with the intended tissue

shapes.

8.3 Results and Discussion

To our surprise, the model trained on the synthetic dataset performed reasonably well, achieving a best average Spearman’s rank correlation of 36.2% on Split 1, where all slices from a single donor were held out as the test set. Even on Split 2, which featured entirely unseen tissue shapes in the test set, the model maintained good performance with a correlation of 31.2%. These results challenged our initial assumption and demonstrated that the LUNA model is inherently capable of generalizing across diverse tissue geometries, provided it is exposed to sufficiently varied examples during training.

This raised a natural question - do we really need the shape guided post-processing at all? To answer this, we closely investigated the results.

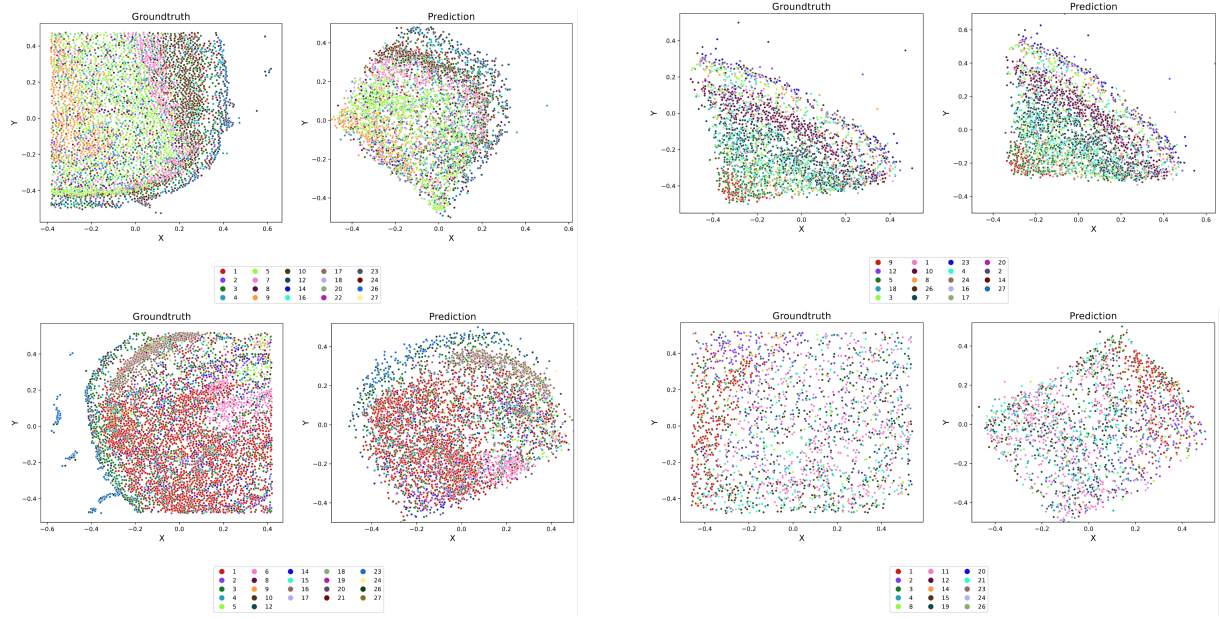


Figure 13: Randomly sampled Examples for synthetic data, split 1

Figures 13 and 14 show some randomly sampled predictions for split 1 and split 2 respectively. While both illustrate good performance, We can notice that for split 2, where the model has never seen the test shapes during training, the predicted tissue shapes are less accurate than in Split 1. This indicates that although the model shows promising generalization, it struggles to capture finer structural nuances of tissue shapes.

This insight reaffirmed the potential value of the shape-guidance pipeline. It can serve as a valuable refinement step in scenarios where the model’s predictions are reasonable but not geometrically faithful, especially in cases of limited dataset diversity or other training constraints. Therefore, we applied our shape-guidance pipeline to the predictions from Split 2 to further investigate its effectiveness.

In Figure 15, we present a representative tissue slice exhibiting a high shape discrepancy between the predicted and ground-truth cell distributions. We apply the shape-guided post-processing pipeline to this example, as illustrated in Figure 16. The pipeline first performs rigid alignment of the predicted coordinates, followed by optimal transport to map them onto a set of sampled points derived from the tissue image.

As evident from the final results, the post-processed predictions align closely with the ground truth, effectively capturing both the tissue shape and underlying gene expression patterns. Notably, the average Spearman’s rank correlation improves from 28.7% (original prediction) to 31.1% after shape-guidance. Furthermore, the cell-type assignments remain consistent, demonstrating that the pipeline preserves biological meaning while enhancing spatial accuracy.

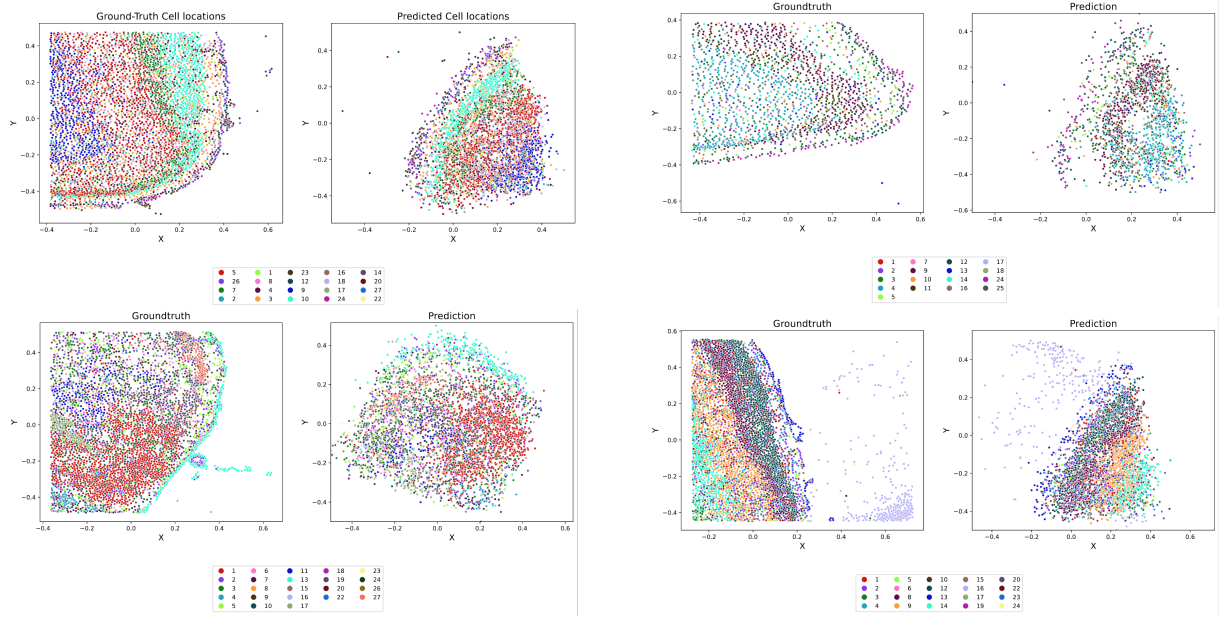


Figure 14: Randomly sampled Examples for synthetic data, split 2 (unseen test shapes)

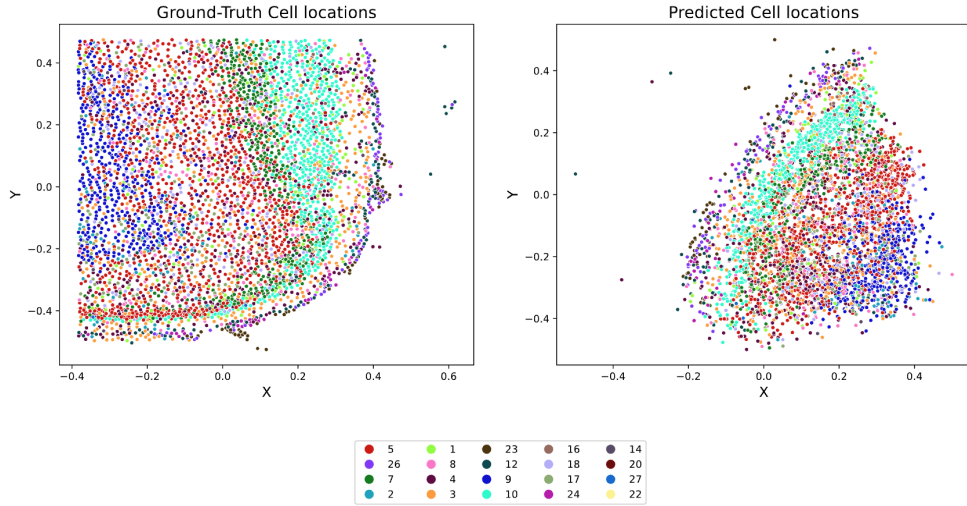


Figure 15: Test slice with a high shape discrepancy

9 Conclusion

We successfully extended the LUNA model by incorporating morphology in two complementary ways: through individual cell morphology and by leveraging global tissue morphology. While the integration of single-cell morphological features led to only marginal performance improvements on our dataset, it serves as a proof of concept for the feasibility of multi-modal models. Such an approach holds strong potential in settings where gene expression data is limited or noisy, and rich morphological data is available.

Additionally, our shape-guided post-processing pipeline produced promising results. In scenarios involving small or non-diverse training datasets, this technique can be particularly valuable for enforcing biologically realistic tissue geometry in the predicted cell layouts. Together, these findings highlight both the challenges and opportunities of integrating morphology into spatial transcriptomic models, and point toward meaningful directions for future work.

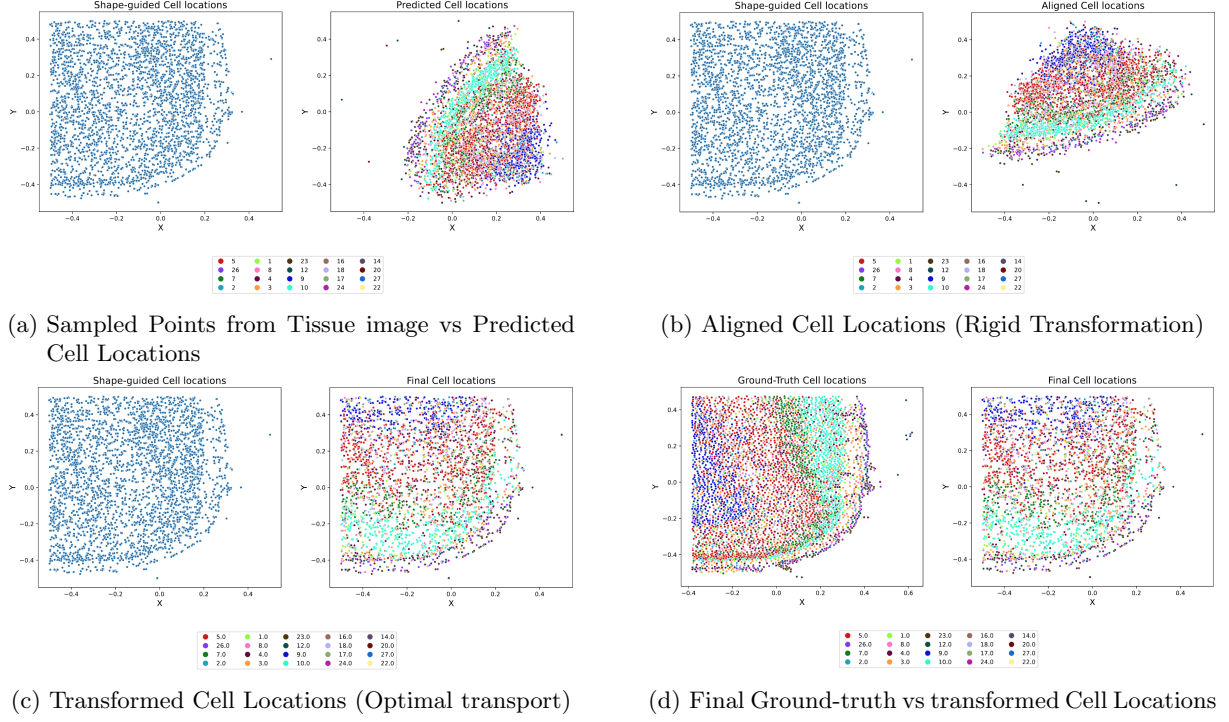


Figure 16: Shape Guidance Pipeline

10 Next Steps

Building on the insights from our analysis, there remains substantial potential for future work. One immediate direction is to explore a broader range of datasets to identify scenarios where integrating morphology provides meaningful improvements.

In particular, we have identified datasets such as human breast tissue, where morphological structures appear more structured and repetitive. Such patterns could contribute significantly to understanding the underlying transcriptomic landscape. Hence, as a next step, we plan to train the LUNA model on the human breast dataset and evaluate whether our shape-guidance approach can better preserve tissue geometry and improve spatial prediction. This would help validate the utility of morphology-guided modeling in biologically complex and morphologically rich tissues.

We also aim to analyze the internal workings of the model by visualizing attention maps and inspecting the learned morphology and gene expression embeddings in the latent space. These analyses could provide deeper insight into how the model reasons over multi-modal inputs and help identify biologically meaningful patterns.

Due to limited time, we were unable to pursue several valuable lines of investigation that could further strengthen the study. Hence additional experiments with alternative train-test splits, varied hyper-parameters, and ablation studies could be done to better understand the robustness and generalizability of the approach. Exploring these would help assess the consistency of the model across different scenarios and parameter settings.

11 Acknowledgments

This project has been a truly enriching and rewarding experience. While understanding machine learning theory is one aspect, applying it to real-world problems presents an entirely different set of challenges and valuable lessons. Through this project, I learned how to develop models from scratch, build robust pipelines for development and monitoring, and apply advanced mathematical concepts such as optimal transport. More importantly, I gained insights into creative problem solving and how to formulate and iterate on ideas in a research setting. I am proud to have contributed to this work, even if in a small way, and I believe there remains much potential to build upon and explore further.

I am extremely grateful for the opportunity to work on such a challenging and impactful project under the guidance

of the MLBio lab. I would like to sincerely thank my mentor, Yist Yu, for her continuous support, insightful discussions, and prompt assistance throughout the semester. I am also deeply thankful to my supervisor, Prof. Maria Brbić, for offering me this opportunity and for her valuable feedback that greatly shaped the direction of the project. I would also like to extend my appreciation to Ramon and Shuo for their helpful discussions. Lastly, the weekly team meetings and intellectually stimulating talks within the lab community have been immensely inspiring.

I look forward to continuing the project and contributing to its future progress in a meaningful direction.

References

- [1] E. Chelebian, C. Avenel, and C. Wählby. Combining spatial transcriptomics with tissue morphology. *Nature Communications*, 16:4452, 2025. doi: 10.1038/s41467-025-58989-8. URL <https://doi.org/10.1038/s41467-025-58989-8>.
- [2] Ao Chen, Sha Liao, Mengnan Cheng, Kailong Ma, Liang Wu, Yiwei Lai, Xiaojie Qiu, Jin Yang, Jiangshan Xu, Shijie Hao, et al. Spatiotemporal transcriptomic atlas of mouse organogenesis using dna nanoball-patterned arrays. *Cell*, 185(10):1777–1792, 2022.
- [3] Rémi Flamary, Nicolas Courty, Alexandre Gramfort, Medjoun Alaya, Aurélien Boisbunon, Stanislas Chambon, Laetitia Chapel, Alexandre Corenflos, Kilian Fatras, Nicolas Fournier, et al. Pot: Python optimal transport. *Journal of Machine Learning Research*, 22(78):1–8, 2021. URL <https://www.jmlr.org/papers/v22/20-451.html>.
- [4] 10X Genomics. Nucleus and cell segmentation algorithms. <https://www.10xgenomics.com/support/software/xenium-onboard-analysis/latest/algorithms-overview/segmentation>, 2025.
- [5] John C Gower. Generalized procrustes analysis. *Psychometrika*, 40(1):33–51, 1975. doi: 10.1007/BF02291478.
- [6] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners, 2021. URL <https://arxiv.org/abs/2111.06377>.
- [7] Lambda Moses and Lior Pachter. Museum of spatial transcriptomics. *Nature Methods*, 19(5):534–546, 2022.
- [8] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. Dinov2: Learning robust visual features without supervision, 2024. URL <https://arxiv.org/abs/2304.07193>.
- [9] Gabriel Peyré and Marco Cuturi. Computational optimal transport. *Foundations and Trends in Machine Learning*, 11(5-6):355–607, 2019. doi: 10.1561/22000000073.
- [10] Samuel G Rodriques, Robert R Stickels, Aleksandrina Goeva, Carly A Martin, Evan Murray, Charles R Vanderburg, Joshua Welch, Linlin M Chen, Fei Chen, and Evan Z Macosko. Slide-seq: A scalable technology for measuring genome-wide expression at high spatial resolution. *Science*, 363(6434):1463–1467, 2019.
- [11] Sergio Marco Salas, Paulo Czarnewski, Louis B. Kuemmerle, Saga Helgadottir, Christoffer Matsson-Langseth, Sebastian Tismeyer, Christophe Avenel, et al. Optimizing xenium in situ data utility by quality assessment and best practice analysis workflows. *bioRxiv*, 2023. doi: 10.1101/2023.02.13.528102. URL <https://www.biorxiv.org/content/10.1101/2023.02.13.528102>. preprint.
- [12] Jun-Han Su, Pu Zheng, Seon S Kinrot, Bogdan Bintu, and Xiaowei Zhuang. Genome-scale imaging of the 3d organization and transcriptional activity of chromatin. *Cell*, 182(6):1641–1659, 2020.
- [13] L.A. Torres-Mendez, J.C. Ruiz-Suarez, L.E. Sucar, and G. Gomez. Translation, rotation, and scale-invariant object recognition. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, 30(1):125–130, 2000. doi: 10.1109/5326.827484.
- [14] Eva Gracia Villacampa, Ludvig Larsson, Reza Mirzazadeh, Linda Kvastad, Alma Andersson, Annelie Mollbrink, Georgia Kokaraki, Vanessa Monteil, Niklas Schultz, Karin Sofia Appelberg, et al. Genome-wide spatial expression profiling in formalin-fixed tissues. *Cell Genomics*, 1(3), 2021.

- [15] Guiping Wang, Jeffrey R Moffitt, and Xiaowei Zhuang. Multiplexed imaging of high-density libraries of rnas with merfish and expansion microscopy. *Scientific reports*, 8(1):4847, 2018.
- [16] Xiao Wang, William E Allen, Matthew A Wright, Emily L Sylwestrak, Nikolay Samusik, Sam Vesuna, Kathryn Evans, Cindy Liu, Charu Ramakrishnan, Jia Liu, et al. Three-dimensional intact-tissue sequencing of single-cell transcriptional states. *Science*, 361(6400):eaat5691, 2018.
- [17] Tingyang Yu, Chanakya Ekbote, Nikita Morozov, Jiashuo Fan, Pascal Frossard, Stéphane d’Ascoli, and Maria Brbić. Tissue reassembly with generative ai. *bioRxiv*, 2025. doi: 10.1101/2025.02.13.638045. URL <https://doi.org/10.1101/2025.02.13.638045>.